

Exact partial structure factors for polydisperse hard-sphere–Yukawa and sticky-hard-sphere fluids, with an assessment of the approximations used in SASfit

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Abstract

We report a complete, validated implementation of the analytic mean-spherical approximation (MSA) solution for an N -component hard-sphere–Yukawa fluid, following Blum & Hernando [J. Phys.: Condens. Matter **14**, 11933 (2002)], together with the resulting exact partial structure factors $S_{ij}(Q)$ and a small-angle-scattering layer that maps a continuous size distribution onto the model and produces the exact scattered intensity. Establishing the implementation required correcting *three* misprinted equations and recognising *one* systematic reading trap; a fourth printed equation is internally inconsistent but is never needed. All three misprints are invisible for a single component with a common coupling, and therefore silently corrupt every polydisperse calculation while leaving every single-component test passing. Each correction is derived from other equations of the same paper rather than fitted, and the final result is verified against three independent references: SASfit’s compiled one-Yukawa library at $N = 1$ (agreement to $\sim 10^{-14}$ in the contact value and $\sim 10^{-10}$ in $S(Q)$), the exact Percus–Yevick hard-sphere and Lebowitz mixture limits as $K \rightarrow 0$, and a purpose-written numerical Ornstein–Zernike solver at $N > 1$ that shares no equation with the analytic route.

We then asked whether these corrections were already known, and found that two of the three are: Blum’s own later paper with Arias [arXiv:cond-mat/0602477 (2006)] prints both of them in corrected form, with no erratum, no footnote and no remark that anything had changed. Our corrections, obtained by probing the paper’s own recursion rather than by transcribing anything, agree with that later paper to 10^{-15} while differing from the 2002 text by up to 18%. The third correction is likewise a silent fix in the same 2006 paper. This is a cautionary result rather than a reassuring one: the corrected equations exist in the literature, but only for a reader who thinks to distrust the 2002 paper and to consult a preprint four years later that never says why it differs. The sequel in turn carries a misprint of its own, which our numerics locate.

The last of those corrections — a single index in the definition of D_{ij} — also removes what had appeared to be the main obstacle to extending the theory: with it, charge (coupling) polydispersity $\delta_i \neq 1$ closes in *closed form* rather than requiring a separate derivation, and polydisperse RMSA, which maps onto exactly that case, becomes analytic. We implement RMSA with a common rescaling factor chosen so that all contact values are non-negative, and verify it against the independent numerical solver to 0.006% in the rescaling factor.

With an exact S_{ij} available, we then quantify — for the first time, as far as we are aware — the error of all six approximate schemes that SASfit offers for combining a structure factor with a size distribution. Two results are of practical consequence. First, the “partial structure factor” option is statistically indistinguishable from the decoupling approximation in every case tested, so the double integral it requires buys nothing. Second, the ranking of the remaining schemes is *not* universal: the local monodisperse approximation is the best choice for repulsive interactions, but for attractive ones the scaling approximation of Gazzillo *et al.* is better by roughly a factor of four. The common rule of thumb favouring the local monodisperse approximation is therefore only half right.

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1 Introduction and scope

Small-angle scattering from a concentrated dispersion measures the product of particle form and inter-particle structure. For a *monodisperse* system this factorises cleanly into $P(Q)$ and $S(Q)$. For a polydisperse system it does not: different size classes have different form factors *and* different mutual correlations, and the intensity becomes a double sum over partial structure factors,

$$I(Q) = \sum_{ij} \sqrt{n_i n_j} F_i(Q) F_j(Q) S_{ij}^{\text{AL}}(Q), \quad (1)$$

with $S_{ij}^{\text{AL}} = \delta_{ij} + \sqrt{n_i n_j} \tilde{h}_{ij}(Q)$ the Ashcroft–Langreth partials and n_i the number density of class i .

Because S_{ij} has rarely been available in closed form for a realistic interaction, scattering analysis has relied on approximations that collapse (1) back into something factorisable. SASfit implements six of them. This report removes the need for that approximation in the hard-sphere–Yukawa case, and then uses the exact result to measure how much each approximation costs.

Scope and restrictions. Throughout, the Yukawa coupling is *factored* with $\delta_i = 1$: a single screening length and a common contact strength shared by all species, with arbitrary polydispersity in *size*. Charge (coupling) polydispersity, $\delta_i \neq 1$, is not covered — see §12. The number of Yukawa terms is $M = 1$ throughout. The sticky-hard-sphere comparison uses SASfit’s own Robertus engine, which likewise takes the stickiness τ to be size-independent.

Provenance. Per the repository’s `AI_USAGE.md`, the derivations in §3 were prepared as scaffolding and have been independently verified by the author before being relied upon; this is stated per item in that section. Every number quoted in this report is regenerated by `report_numbers.py` rather than transcribed.

2 Theoretical background

2.1 Model and closure

We take N species with hard-core diameters σ_i and number densities ρ_i , interacting through a hard core plus a single Yukawa tail. Outside contact the MSA closure fixes the direct correlation function (Blum & Hernando, hereafter BH02, eq. (5)),

$$c_{ij}(r) = \frac{K_{ij} e^{-z(r-\sigma_{ij})}}{r}, \quad r > \sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j), \quad (2)$$

while the hard core fixes the pair correlation inside it, $h_{ij}(r) = -1$ for $r \leq \sigma_{ij}$ (eq. (6)). In the factored case $K_{ij} = K\delta_{ij}$; with $\delta_i = 1$ this is simply $K_{ij} = K$. Because the MSA sets $c = -\beta U$ outside the core, the sign convention is the opposite of what the form of (2) might suggest: a screened-Coulomb *repulsion* has $U > 0$ and therefore $K < 0$, while $K > 0$ describes an *attraction*. This is confirmed by both independent indicators — at $\varphi = 0.1$, $K = +1$ gives $g(\sigma^+) = 2.13$ and $S(0) = 0.636$, against the hard-sphere values 1.296 and 0.456 (contact enhanced, compressibility raised: attraction), while $K = -1$ gives 0.504 and 0.348 (contact depleted, compressibility lowered: repulsion).

2.2 Baxter–Wertheim factorisation

The Ornstein–Zernike equation is solved by the Baxter–Wertheim factorisation (BH02 eq. (7), eq. (12)), which writes $\mathbf{I} - \rho\hat{\mathbf{C}}(k)$ as a product of a matrix analytic in the upper half k -plane and its conjugate. The factor function $q_{ij}(r)$ has the piecewise form (BH02 eq. (17), eq. (18))

$$q_{ij}(r) = q_{ij}^0(r) + D_{ij}^{(1)} e^{-zr}, \quad \lambda_{ji} < r, \quad (3)$$

$$q_{ij}^0(r) = \frac{1}{2} A_j \left[\left(r - \frac{\sigma_j}{2} \right)^2 - \left(\frac{\sigma_i}{2} \right)^2 \right] + \beta_j \left[\left(r - \frac{\sigma_j}{2} \right) - \frac{\sigma_i}{2} \right] + C_{ij}^{(1)} e^{-z\sigma_j/2} \left[e^{-z(r-\sigma_j/2)} - e^{-z\sigma_i/2} \right], \quad \lambda_{ji} < r < \sigma_{ij}, \quad (4)$$

with $\lambda_{ji} = \frac{1}{2}(\sigma_j - \sigma_i)$. The unknowns are the Baxter amplitudes a_j and the auxiliary quantities A_j , β_j , C_{ij} , D_{ij} and $\hat{B}_j$.

2.3 The Blum–Hernando solution

The purely geometric quantities are

$$\zeta_k = \sum_i \rho_i \sigma_i^k, \quad \Delta = 1 - \frac{\pi}{6} \zeta_3, \quad A_j^0 = \frac{2\pi}{\Delta} \left[1 + \frac{1}{2} \zeta_2 \frac{\pi}{\Delta} \sigma_j \right], \quad \beta_j^0 = \frac{\pi}{\Delta} \sigma_j, \quad (5)$$

and the solution is built from them by (BH02 eq. (23), eq. (24))

$$A_j = A_j^0 + \frac{\pi}{\Delta} a_j P, \quad \beta_j = \beta_j^0 + a_j \Delta^{(1)}. \quad (6)$$

Three auxiliary functions recur throughout,

$$\phi_0(x) = \frac{1 - e^{-x}}{x}, \quad \phi_1(x) = \frac{1 - x - e^{-x}}{x^2}, \quad \psi_1(x) = \frac{1 - \frac{x}{2} - (1 + \frac{x}{2})e^{-x}}{x^3}, \quad (7)$$

all entire, and satisfying $\phi_1 = x\psi_1 - \phi_0/2$ (BH02 eq. (45)).

Two further vectors, Π_j and X_j , carry the dependence on $\hat{B}$. BH02 gives them twice: once through matrices $\hat{\mathcal{X}}, \hat{\mathcal{J}}$ (eq. (34)–eq. (39)), and once explicitly (eq. (30), eq. (32), eq. (33)),

$$\Delta^{(1)} = -\frac{2\pi}{z^2 \Delta} \sum_{\ell} \rho_{\ell} \left(1 + \frac{z\sigma_{\ell}}{2} \right) \delta_{\ell} - \frac{2\pi}{\Delta} \sum_{\ell} \rho_{\ell} \sigma_{\ell}^3 \psi_1(z\sigma_{\ell}) \hat{B}_{\ell}, \quad (8)$$

$$X_i = \delta_i + \sigma_i \phi_0(z\sigma_i) \hat{B}_i + \sigma_i \Delta^{(1)}, \quad (9)$$

$$\Pi_j = \hat{B}_j + \left(1 + \frac{\sigma_j z}{2} \right) \Delta^{(1)} + \frac{\sigma_j}{2} \sum_{\ell} \rho_{\ell} \beta_{\ell}^0 X_{\ell}. \quad (10)$$

That redundancy is what makes the errata of §3 detectable.

2.4 Closing the system

Two equivalent routes close the problem. The first solves the primary MSA closure (BH02 eq. (55)) for the N amplitudes a_j , with $\hat{B}$ obtained from its own linear self-consistency (eq. (62)–eq. (64)). The second uses the scaling structure of §4 of BH02: eq. (72) states $\Pi_i = -\Gamma X_i$ with Γ a scalar at $M = 1$, which is linear in $\hat{B}$ at fixed Γ , and eq. (100b),

$$2\pi K = -\frac{2\Gamma(z + \Gamma)}{D_2}, \quad D_2 = \sum_k \rho_k X_k^2, \quad (11)$$

supplies the one remaining scalar equation, with the amplitudes then *derived* as $a_j = 2\Gamma X_j / D_2$. The whole problem reduces to a one-dimensional root find in Γ .

Both are implemented (`solve()` and `solve_gamma()`) and, once the errata below are applied, they agree to machine precision. They are retained as mutual cross-checks: they carry different unknowns, so agreement is informative rather than tautological. The Γ route additionally handles $K = 0$ exactly, where the other degenerates.

3 Errata in the published equations

Five separate obstacles had to be cleared before the implementation closed. They are not all of the same kind, and it matters to keep them apart — a distinction earlier drafts of this report did not draw sharply enough. *Three* are genuine misprints: an equation as printed says something the authors demonstrably did not mean, and the corrected form is forced by other equations of the same paper. *One* is an internal inconsistency in an equation we do not need. And *one* — the one that cost the most time — is not an error in the paper at all, but a trap for the implementer: an equation that is entirely correct for what it states, and that states something other than what a reader building a structure-factor code is looking for.

The three misprints share a property that explains how they survived peer review and twenty-four years of citation: each is invisible in the simplest case. Two of them collapse to the correct expression whenever $N = 1$, because the two interchanged indices then take the same value; the third collapses whenever all δ_i are equal, which is the case in every application that does not model charge polydispersity. A single-component validation against a reference implementation — exactly the validation an implementer would naturally run first, and exactly the one we ran — passes at machine precision with all three misprints still in place. This is worth stating plainly because it undercuts a tempting inference: agreement with published single-component numbers is *not* evidence that the multicomponent equations have been transcribed correctly.

Table 1 summarises the five items. Each correction is derived below from *other* equations of the same paper, never fitted to our numerics; §4 then asks, separately, whether each was already known.

Table 1: The five items, classified. “Simplest case” indicates whether the item has any effect on a single-component calculation with a common coupling $\delta_i \equiv 1$ — that is, whether a monodisperse validation can see it. The last column anticipates §4.

Equation	Kind	Nature	Visible in simplest case	Already in the literature?
BH02 eq. (35)	misprint	$j \leftrightarrow \ell$ interchange	no	yes, BA eq. (76)
BH02 eq. (38)	misprint	$j \leftrightarrow \ell$ interchange	no	yes, BA eq. (76)
BH02 eq. (22)	misprint	δ on the column index	no	yes, BA eq. (25)
BH02 eq. (31)	inconsistent	disagrees with eq. (27), eq. (28)	yes	not addressed
BH02 eq. (20)	<i>correct</i>	is q' , not the contact value	—	yes, BA eq. (26)

Table 2: Cost of leaving each item uncorrected, measured on the three-component test system of §8.4 ($\sigma = (1, 1.5, 2.5)$, $z = 4$, $K = 0.8$).

Equation	Effect if uncorrected
BH02 eq. (35)	$\sim 2 \times 10^{-2}$ in $\Pi^{(n)}$; up to 18% in $\hat{\mathcal{I}}$ at $z\sigma \sim 2$
BH02 eq. (38)	$\sim 10^{-2}$ in $X^{(n)}$
BH02 eq. (22)	$g_{ij} \neq g_{ji}$ by 14–35% once the δ_i differ
BH02 eq. (31)	$O(1)$ in $P^{(n)}$ — but $P^{(n)}$ is taken from eq. (27)/eq. (28) instead
BH02 eq. (20)	$> 40\%$ in $g(\sigma^+)$ at large z , if misread as the contact value

3.1 The contact value is a jump, not a derivative

This is the item that is not an erratum, and it is worth being precise about why, because the report’s earlier drafts got the classification wrong.

BH02 eq. (20) reads, verbatim,

$$q'_{ij}(\sigma_{ji}) = A_j(\sigma_i/2) + \beta_j - \sum_{m=1}^M z_m (C_{ij}^{(m)} + D_{ij}^{(m)}) e^{-z_m \sigma_{ji}}, \quad (12)$$

and the sentence immediately following it says that it “will be used below in connection with the symmetry requirements for the factor functions”. That is exactly what it is for. Equation (12) is the derivative of the *full* factor function of eq. (17) — core part plus Yukawa tail — evaluated on one side of the core, and it is correct as printed.

It is not the contact value. The contact value is the *jump* in q' across $r = \sigma_{ji}$, and only the core-region part $q^{0'}$ of eq. (18) survives the difference. Blum & Arias state this explicitly four years later, writing the contact value as $2\pi\sigma_{ij}g_{ij}(\sigma_{ij}) = q'_{ij}(\sigma_{ji}^-) - q'_{ij}(\sigma_{ji}^+)$ — the two-sided difference that BH02 leaves implicit. Take (12) for the contact value and the D term rides along spuriously, costing more than 40% in $g(\sigma^+)$ at large z . Nothing in BH02 licenses that misreading; nothing in BH02 warns against it either. The correct statement is

$$2\pi\sigma_{ij}g_{ij}(\sigma_{ij}^+) = A_j \frac{\sigma_i}{2} + \beta_j - \sum_m z_m C_{ij}^{(m)} e^{-z_m \sigma_{ij}} \quad (13)$$

— with C only. The distinction is invisible as $K \rightarrow 0$, where $C, D \rightarrow 0$.

This was established from the large- s asymptotics of the Laplace transform $\tilde{g}(s) = \mathcal{L}[rg(r)]$, which the compiled reference library computes directly. Writing $\sigma g(\sigma^+) = \lim_{s \rightarrow \infty} s e^{s\sigma} \tilde{g}(s)$ and inserting the library’s own expression gives $g(\sigma^+) = (a + b) - cZe^{-Z}$ exactly in its Baxter coefficients, which maps onto (13) term by term.

Eliminating C_{ij} with eq. (21) and eq. (22), using the symmetry of $\tilde{g}$ (valid for $\delta_i = 1$), yields the implementable form

$$2\pi\sigma_{ij}g_{ij}(\sigma_{ij}^+) = A_j \frac{\sigma_i}{2} + \beta_j - z a_j + a_j \hat{B}_i e^{-z\sigma_i}. \quad (14)$$

A correction to our own earlier reading. An earlier draft of this report recorded that Blum & Arias eq. (26) “carries a spurious factor $e^{-z\sigma_{ij}}$ on the $z a_j$ term”. That was wrong, and the error was ours, not theirs. BA eq. (26) reads

$$2\pi\sigma_{ij}g_{ij}(\sigma_{ij}) = A_j(\sigma_i/2) + \beta_j - \sum_m (z_m \delta_i^{(m)} - \mathcal{B}_i^{(m)}) a_j^{(m)} e^{-z_m \sigma_{ij}}, \quad (15)$$

in a scaling convention that differs from ours by species-dependent exponentials — BA eq. (4) defines $\delta_i = d_i e^{-z_n \sigma_i/2}$, and the amplitudes are scaled to match. Under the map

$$\delta_i^{\text{BA}} = \delta_i e^{+z\sigma_i/2}, \quad a_j^{\text{BA}} = a_j e^{+z\sigma_j/2}, \quad \mathcal{B}_i^{\text{BA}} = \hat{B}_i e^{-z\sigma_i/2}, \quad (16)$$

the exponentials cancel identically: $z\delta_i^{\text{BA}} a_j^{\text{BA}} e^{-z\sigma_{ij}} = z\delta_i a_j$ and $\mathcal{B}_i^{\text{BA}} a_j^{\text{BA}} e^{-z\sigma_{ij}} = \hat{B}_i a_j e^{-z\sigma_i}$, which is (14) term for term. Evaluated numerically, BA eq. (26) under (16) reproduces our contact matrix to 0, 4.4×10^{-16} and 2.2×10^{-16} at $N = 1, 2, 3$ — the last with unequal δ_i . So (15) is not merely compatible with our result: it *is* our result, published in 2006. We record the earlier misattribution because a report that lists other people’s misprints owes the reader its own.

Independent confirmation. BH02 eq. (100), on a later page, gives a second contact-value expression routed through Γ and X_i rather than through A, β, C : $2\pi\sigma_{ij}[g_{ij} - g_{ij}^0] = -(z + \Gamma)X_i a_j$. At $N = 1$ it reproduces (14) with relative error exactly zero.

3.2 $\hat{I}$ and $\hat{\mathcal{J}}$: a $j \leftrightarrow \ell$ interchange

Substituting (8) into (9) and matching the form $X_j = \hat{\gamma}_j + \sum_\ell \hat{\mathcal{J}}_{j\ell} \hat{B}_\ell$ (eq. (37)) gives

$$\hat{\mathcal{J}}_{j\ell} = \delta_{j\ell} \sigma_j \phi_0(z\sigma_j) - 2\rho_\ell \beta_j^0 \sigma_\ell^3 \psi_1(z\sigma_\ell), \quad (17)$$

whereas eq. (38) prints $-2\rho_\ell\beta_\ell^0\sigma_j^3\psi_1(z\sigma_j)$: the indices j and ℓ are interchanged in the second term. The two coincide at $N = 1$.

The decisive point is that the *same* substitution reproduces $\hat{\gamma}_j$ of eq. (39) and $\hat{\xi}_j$ of eq. (36) exactly as printed. The two vectors are therefore correct and only the two matrices carry the interchange — corroboration from the paper’s own equations, not a fit. Numerically, with random $\hat{B}$, the derived $\hat{\mathcal{J}}$ reproduces the explicit route to machine precision while the printed one does not (Table 3).

Table 3: Maximum $|X - X_{\text{explicit}}|$ over random $\hat{B}$ vectors, using the derived versus the printed $\hat{\mathcal{J}}$, against the explicit route (8)–(9).

N	derived	as printed
1	1.1×10^{-16}	1.1×10^{-16}
2	1.1×10^{-16}	3.0×10^{-3}
3	1.1×10^{-16}	1.1×10^{-2}

Rather than hand-transcribe the corrected matrices, the implementation obtains $\hat{\mathcal{I}}, \hat{\mathcal{J}}, \hat{\xi}, \hat{\gamma}$ by *exactly probing* the affine map $\hat{B} \mapsto (\Pi, X)$ of (8)–(10) with the standard basis. This is transcription-free: no printed matrix is relied upon at all.

3.3 $P^{(n)}$: eq. (31) is inconsistent

BH02 gives three expressions for $P^{(n)}$. Equations eq. (27) and eq. (28) agree with each other to machine precision at every N ; eq. (31), $P = \sum_\ell \rho_\ell \beta_\ell^0 X_\ell - z\Delta^{(n)}$, does not, already at $N = 1$. The implementation uses eq. (27) and does not use eq. (31).

3.4 D_{ij} : δ carries the row index

BH02 eq. (22) prints $D_{ij}^{(n)} = -\delta_j^{(n)} a_j^{(n)} e^{z_n \sigma_{ij}}$. The coupling amplitude must instead carry the *row* index,

$$D_{ij}^{(n)} = -\delta_i^{(n)} a_j^{(n)} e^{z_n \sigma_{ij}} \quad (18)$$

This is invisible whenever $\delta_i = 1$ — which is every calculation in §8 and §10 — and so cannot be detected by any common-coupling test, however stringent.

The evidence is a physical requirement rather than a numerical preference. The contact matrix must satisfy $g_{ij} = g_{ji}$, since $g_{ij}(r)$ is a pair correlation function; nothing in the assembly imposes it. With eq. (22) as printed and $\delta_i \neq 1$, the computed contact matrix comes out asymmetric by $|g - g^T| \approx 0.26\text{--}0.35$, and disagrees with the independent numerical OZ solver by 15–21%. With (18) the asymmetry falls to $\sim 5 \times 10^{-6}$ and the agreement reaches that solver’s own accuracy floor ($\sim 2.6 \times 10^{-3}$ at the grid used), identical to what it achieves at $\delta_i = 1$. The correction was located by reverse-engineering the D_{ij} implied by a numerically exact g_{ij} and $\tilde{g}_{ij}$ through eq. (21), then testing candidate index placements: of the four, only (18) comes within 2% of the implied matrix, the others missing by 42–230%.

It does not affect any result reported elsewhere here. At $\delta_i = 1$ the two forms of D_{ij} are literally identical. This was verified rather than assumed: every quantity in §8 and §10 is unchanged — $N = 1$ contact 6.36×10^{-14} , $N = 1$ $S(Q)$ 2.66×10^{-10} , the $K \rightarrow 0$ limit 7.98×10^{-11} , all six $N > 1$ conditions still at machine precision with bit-identical tail amplitudes, and the approximation table of §10 reproduced digit for digit.

3.5 The rank-one form of Blum & Arias

Blum & Arias eq. (28), eq. (29) offer an attractive route to the k -space matrix as a sum of $M + 2$ rank-one terms, with eq. (34) collapsing its determinant to an $(M+2) \times (M+2)$ one. Implemented as printed, the rank-reduction identity $\det \mathbf{M} \equiv D_\tau$ holds exactly — so the vectors are at least self-consistent — but the resulting $\mathbf{M}$ is wrong by a large, s -dependent factor, growing from 6.4 at $s = 0.5$ to 284 at $s = 10$. A term-by-term comparison against BH02 eq. (52) shows the third term carrying $e^{-s\sigma_i/2}$ where eq. (52) has $e^{-s\sigma_{ij}}$. Whether this is a misreading or a typographical error was not resolved: the route was abandoned once the BH02 eq. (52)/eq. (53) route, which reproduces the reference $\tilde{g}(s)$ to all displayed digits at every s , was available.

4 Status of these corrections in the literature

A correction derived from first principles and a correction already known are different things, and it would be poor practice to present the first as though it were novel without checking for the second. The multicomponent hard-sphere Yukawa MSA has been solved repeatedly since 1977, so any misprint in one account of it might well have been noticed — and quietly fixed — somewhere else. This section asks that question for each item in Table 1, and answers it.

The short answer is that *two of the three misprints, and the reading trap as well, are already corrected in the literature* — in Blum’s own sequel with Arias [2], four years later, with no erratum, no footnote, and no statement anywhere in that paper that the earlier equations had been changed.

4.1 Method

Rather than compare prose, we compared equations, and then compared numbers. For each item we (i) read the relevant equation of BH02 verbatim from the published page, (ii) read the corresponding equation of BA verbatim from the arXiv version, and (iii) evaluated *both* against our own construction, which is transcription-free: the solver never transcribes eq. (35) or eq. (38), but recovers the matrices $\hat{\mathcal{I}}, \hat{\mathcal{J}}$ by probing the affine map $\hat{B} \mapsto (\Pi, X)$ of eq. (30), eq. (32), eq. (33) with the standard basis (§3.2). Step (iii) matters: it means the comparison is not our reading of one paper against our reading of another, but two independent readings against a construction that reads neither.

4.2 $\hat{\mathcal{I}}$ and $\hat{\mathcal{J}}$: corrected in BA eq. (76)

The two papers print the same pair of matrices. BH02 eq. (35) and eq. (38) read

$$\hat{\mathcal{I}}_{j\ell}^{(n)} = \delta_{j\ell} + \rho_\ell \left[\beta_\ell^0 \frac{\sigma_j^2}{2} \phi_0(z_n \sigma_j) - (A_\ell^0 + z_n \beta_\ell^0) \sigma_j^3 \psi_1(z_n \sigma_j) \right], \quad (19)$$

$$\hat{\mathcal{J}}_{j\ell}^{(n)} = \delta_{j\ell} \sigma_j \phi_0(z_n \sigma_\ell) - 2\rho_\ell \beta_\ell^0 \sigma_j^3 \psi_1(z_n \sigma_j), \quad (20)$$

whereas BA eq. (76) reads

$$\hat{\mathcal{I}}_{j\ell}^{(n)} = \delta_{j\ell} - \rho_\ell \sigma_\ell^2 [\beta_j^0 \phi_1(z_n \sigma_\ell) + A_j^0 \sigma_\ell \psi_1(z_n \sigma_\ell)], \quad (21)$$

$$\hat{\mathcal{J}}_{j\ell}^{(n)} = \delta_{j\ell} \sigma_j \phi_0(z_n \sigma_\ell) - 2\rho_\ell \beta_j^0 \sigma_\ell^3 \psi_1(z_n \sigma_\ell). \quad (22)$$

The indices set in bold are the ones that differ: everywhere BH02 carries ℓ on an amplitude and j on a diameter, BA carries j on the amplitude and ℓ on the diameter. Three things are *not* in bold, and their agreement is as informative as the disagreement. The density weight ρ_ℓ — the one index that is genuinely a summation weight rather than a label — sits on ℓ in both. The Kronecker terms agree, and in any case cannot differ, since $\delta_{j\ell}$ forces $j = \ell$. And the argument of ϕ_0 in $\hat{\mathcal{J}}$ is $z_n \sigma_\ell$ in *both* papers, so that one argument is not interchanged either. What differs is

exactly the set of subscripts that a compositor working from a handwritten j and a handwritten ℓ would be likely to confuse.¹

That the two are otherwise the *same* equation, rather than two different derivations, can be shown outright. BA eq. (24) prints the identity

$$\phi_1(x) = x\psi_1(x) - \frac{1}{2}\phi_0(x), \quad (23)$$

which our implementation satisfies to 2.2×10^{-16} over $x \in [10^{-3}, 40]$. Interchanging $j \leftrightarrow \ell$ in the bracket of (19) and applying (23),

$$\begin{aligned} & \beta_j^0 \frac{\sigma_\ell^2}{2} \phi_0(z\sigma_\ell) - (A_j^0 + z\beta_j^0) \sigma_\ell^3 \psi_1(z\sigma_\ell) \\ &= -\beta_j^0 \sigma_\ell^2 \left[z\sigma_\ell \psi_1(z\sigma_\ell) - \frac{1}{2}\phi_0(z\sigma_\ell) \right] - A_j^0 \sigma_\ell^3 \psi_1(z\sigma_\ell) = -\sigma_\ell^2 [\beta_j^0 \phi_1(z\sigma_\ell) + A_j^0 \sigma_\ell \psi_1(z\sigma_\ell)], \end{aligned} \quad (24)$$

which is precisely the bracket of (21). So eq. (35) and (21) are algebraically identical up to the index interchange. That is the signature of a transcription slip, not of a revised derivation.

There is no ambiguity about whether BA intends (21)–(22) as a restatement of BH02 rather than as a fresh derivation. BA’s Appendix I opens with the sentence “We quote the results from the review of Blum and Hernando [34]”, and the block containing eq. (76) is introduced with “Here we define [34]” — reference [34] being BH02 itself. The authors are therefore explicitly reproducing the 2002 equations, and what they reproduce differs from what 2002 printed, with no remark anywhere that anything has changed. That is what makes this a silent correction rather than an independent result that happens to disagree.²

Table 4 settles which one is right. The solver’s transcription-free probe agrees with BA to machine precision at every N and every screening length tested, and disagrees with BH02 by between 1.1% and 18% as soon as $N > 1$. At $N = 1$ all three agree exactly, because $j = \ell$ makes the interchange vacuous — which is the whole reason the misprint is survivable.

Table 4: Adjudication of eq. (35)/eq. (38). Entries are $\max |\cdot| / \max |\cdot|$ over both matrices, comparing the solver’s probe of eq. (30)/eq. (32)/eq. (33) — which transcribes neither paper — against each printed version. Generated by `blum_arias_IJ_check.py`.

Case	z	probe vs. BA eq. (76)	probe vs. BH02 eq. (35)/eq. (38)
$N = 1, \sigma = (1)$	2	1.8×10^{-16}	1.8×10^{-16}
$N = 1, \sigma = (1)$	6	2.0×10^{-16}	2.0×10^{-16}
$N = 2, \sigma = (0.8, 1.2)$	2	3.8×10^{-16}	5.8×10^{-2}
$N = 2, \sigma = (0.8, 1.2)$	6	1.1×10^{-15}	1.1×10^{-2}
$N = 3, \sigma = (1, 1.5, 2.5)$	2	4.3×10^{-16}	1.8×10^{-1}
$N = 3, \sigma = (1, 1.5, 2.5)$	6	9.7×10^{-16}	2.5×10^{-2}
$N = 4, \sigma = (0.6, 0.9, 1.3, 1.8)$	2	4.4×10^{-16}	1.2×10^{-1}
$N = 4, \sigma = (0.6, 0.9, 1.3, 1.8)$	6	1.3×10^{-15}	2.1×10^{-2}

The two accompanying *vectors* are unaffected: BH02 eq. (36) and eq. (39) agree with BA eq. (77) and with the probe to $\leq 4.2 \times 10^{-16}$ at every N tested. (BA eq. (77) writes $\hat{\xi}_j$ with $A_j^0 + z_n Q'(\sigma_{\ell j})$ and $\hat{\gamma}_j$ with $2\pi\sigma_j/z_n^2\Delta$ where BH02 writes $z_n\beta_j^0 + A_j^0(1 + z_n\sigma_\ell/2)$ and $2\beta_j^0/z_n^2$; these coincide identically given $\beta_j^0 = \pi\sigma_j/\Delta$.) It is only the two matrices that carry the interchange — which is itself evidence for the transcription-slip reading, since a genuine error in the derivation would not spare the vectors that come out of the same algebra.

¹BH02 eq. (35) is also printed with an unbalanced parenthesis — one closing bracket too many after $\psi_1(z_n\sigma_j)$ — which is at least consistent with the typesetting-accident reading.

²BA’s reference [34] gives the page as 10933 rather than 11933. The theme of this section is not confined to the displayed equations.

4.3 D_{ij} : corrected in BA eq. (25)

BH02 eq. (22) reads, verbatim, $D_{ij}^{(n)} = -\delta_j^{(n)} a_j^{(n)} e^{z_n \sigma_{ij}}$, while BA eq. (25) reads $D_{ij}^{(m)} = -\delta_i^{(m)} a_j^{(m)}$ — again in BA’s absorbed scaling (16), and again with the index moved. BA eq. (25) also supplies $C_{ij}^{(m)} = (\delta_i^{(m)} - \mathcal{B}_i^{(m)}/z_m) a_j^{(m)}$ with $\mathcal{B}_i^{(m)} = 2\pi \sum_k \rho_k g_{ik}^{(m)} \delta_k^{(m)}$, i.e. with the δ -weighted contraction over k in place of a bare δ_j — exactly the structure §6 derives.

Two independent arguments confirm the row index without leaving BH02 at all.

First, internal consistency. As printed, eq. (22) is already odd on its face: its left-hand side carries a row index i , but its right-hand side depends on i only through the symmetric σ_{ij} , so all the coupling information sits on the column. Substituting it into BH02’s own eq. (21), $C_{ij} + D_{ij} = (2\pi/z_n) \sum_k \rho_k \tilde{g}_{ik} D_{kj}$, gives $C_{ij} + D_{ij} \propto \delta_j \sum_k \rho_k \tilde{g}_{ik} e^{z \sigma_{kj}}$, whereas the corrected form gives $C_{ij} + D_{ij} = -(2\pi/z) a_j \sum_k \rho_k \tilde{g}_{ik} \delta_k e^{z \sigma_{kj}} = -(\mathcal{B}_i/z) a_j$, which is BA eq. (25)’s $C + D$ identically. Only the corrected version closes eq. (21).

Second, symmetry. The pair distribution function must satisfy $g_{ij} = g_{ji}$ — and this is not an external requirement imposed by us, but one BH02 itself states, as eq. (77) on a later page. The printed eq. (22) therefore contradicts eq. (77) of the same paper. This single requirement decides the index with no external reference whatever, and Table 5 shows the result: with δ on the row index the contact matrix is symmetric to machine precision; with δ on the column index, as printed, it is asymmetric by 14–35% the moment the δ_i differ. When all δ_i are equal the two forms are identical — which is why no calculation that assumes a common coupling can ever detect this misprint.

Table 5: Relative asymmetry $\max |g - g^\top| / \max |g|$ of the contact matrix, built with δ carried by the row index (corrected) and by the column index (as eq. (22) prints it). $z = 3$, $K = 0.6$. Generated by `eq22_symmetry_test.py`.

Case	δ on row i	δ on column j
$N = 2$, $\delta = (1, 1)$ (<i>degenerate</i>)	0	0
$N = 2$, $\delta = (1, 0.7)$	1.2×10^{-16}	1.4×10^{-1}
$N = 2$, $\delta = (1.4, 0.5)$	1.7×10^{-16}	3.5×10^{-1}
$N = 3$, $\delta = (1.3, 1, 0.6)$	8.6×10^{-17}	2.5×10^{-1}
$N = 4$, $\delta = (1.5, 1.1, 0.9, 0.4)$	1.9×10^{-16}	2.5×10^{-1}

4.4 $P^{(n)}$: not addressed anywhere

BH02 eq. (31) disagrees with eq. (27) and eq. (28), which agree with each other at all N (§3.3). BA does not reprint eq. (31) in a comparable form, so we have no external adjudication for it. Since our implementation takes $P^{(n)}$ from eq. (27)/eq. (28), the point is moot in practice; we record it only so that a future reader is not caught by it.

4.5 The sequel has its own misprint

It would be wrong to conclude that BA is simply the corrected version of BH02 and can be trusted wholesale. Comparing the two papers’ expressions for $\Pi_j^{(n)}$, BH02 eq. (32) carries a factor $(1 + \sigma_j z_n/2)$ on the $\Delta^{(n)}$ term that BA eq. (74) does not — BA prints $\Delta^{(n)}$ bare. (BA eq. (74) also writes the third term as $(\pi/2\Delta) \sigma_j P^{(n)}$ where BH02 eq. (32) writes $\frac{1}{2} \sigma_j \sum_\ell \rho_\ell \beta_\ell^0 X_\ell^{(n)}$; these agree only through eq. (31), which is the one BH02 equation we already know to be inconsistent with eq. (27) and eq. (28). The two discrepancies may well be the same discrepancy.) Our probe of eq. (30)/eq. (32)/eq. (33) — which is what everything else in this report is built on, and which reproduces eq. (36) and eq. (39) exactly — validates the BH02 form. Similarly, the rank-one route of BA eq. (28)/eq. (29) (§3.5) does not reproduce BH02 eq. (52) for us. Each paper, in other words, carries transcription damage that the other does not, and the intersection of the two

— checked numerically against a construction that transcribes neither — is what we have actually validated. That is the working conclusion we would offer to anyone implementing this theory: treat both papers as primary sources for the *structure* of the solution and neither as authoritative for any individual index.

4.6 Why no erratum was ever published

The obvious question is how misprints of this consequence survive twenty-four years in a well-cited paper. The reference library assembled for this work suggests an answer: BH02 is not the account that people implement.

The multicomponent hard-sphere Yukawa MSA has an unusually long and branching history. Høye & Blum solved the single-component multi-Yukawa case in 1977 [3]; Blum & Høye extended it to mixtures the following year [4], twenty-four years before BH02, and that is the paper most implementations still cite. Niizeki gave an independent multicomponent solution in 1981 [6]. Ginoza developed the factorisable case — $K_{ij} = K\delta_i\delta_j$, which is exactly our $\delta_i \neq 1$ — through the mid-1980s, and with Yasutomi applied it to colloidal dispersions with simultaneous size and “charge” polydispersity in 1998 [8], using a Schulz size distribution and a scaling parameter promoted from scalar to matrix [7]. Herrera, Blum & García-Llanos treated different-sized spheres in 1996. On the applications side, the two-Yukawa analyses of protein and micellar solutions that motivated BA in the first place use Høye & Blum (1977), not BH02, and handle polydispersity by the decoupling approximation rather than by partial structure factors. Kalyuzhnyi & Hlushak’s polydisperse multi-Yukawa work [9] is thermodynamic rather than structural.

So the specific combination that exposes these misprints — BH02’s own equations, $N > 1$, and structural rather than thermodynamic output — appears to be rare. An implementer working from Blum & Høye (1978) or from Ginoza would never meet eq. (35), eq. (38) or eq. (22) at all. An implementer working from BA meets them already corrected. And an implementer working from BH02 at $N = 1$, as we did first, sees nothing wrong. There is a plausible reading in which the corrections were made silently between 2002 and 2006 precisely because someone — most likely the authors — hit them while preparing the sequel, and simply typeset the right thing the second time.

4.7 What this does and does not tell us about our own validation

One inference must be resisted. Our implementation reproduces SASfit’s compiled one-Yukawa library to $\sim 10^{-14}$ (§8.1), and that library encodes a published, independently derived single-component solution. It is tempting to argue from this that the equations we transcribed must be right, and hence that any misprints must already have been shaken out of the literature. They are unrelated claims. Every one of the three misprints is *exactly* invisible at $N = 1$ with $\delta_i \equiv 1$, which is the only regime in which that comparison is possible. The single-component agreement is strong evidence about the closure, the contact value and the $S(Q)$ assembly; it is no evidence at all about eq. (35), eq. (38) or eq. (22).

What does bear on those is the $N > 1$ numerical solver of §8.4, which shares no equation with the analytic route, and — now — the independent agreement with BA. Those two are what the corrections rest on.

5 Partial structure factors

With q_{ij} known, eq. (53) gives its Laplace transform, and the Baxter–Wertheim factorisation conjugated by $\mathbf{D} = \text{diag}(\sqrt{\rho_i})$ gives a manifestly Hermitian, positive-definite form:

$$m_{ij}(Q) = \delta_{ij} - \sqrt{\rho_i\rho_j} \tilde{q}_{ij}(is) \Big|_{s=-iQ}, \quad \boxed{\mathbf{S}^{\text{AL}}(Q) = [\mathbf{m}(Q) \mathbf{m}(Q)^\dagger]^{-1}} \quad (25)$$

with, for $M = 1$ and $\delta_i = 1$,

$$\begin{aligned}\tilde{q}_{ij}(is) &= e^{-s\lambda_{ji}} \left[\sigma_i^3 \psi_1(s\sigma_i) A_j + \sigma_i^2 \phi_1(s\sigma_i) \beta_j \right. \\ &\quad \left. + \frac{1}{s+z} \left((C_{ij} + D_{ij}) e^{-z\lambda_{ji}} - C_{ij} e^{-z\sigma_{ij}} - z\sigma_i \phi_0(s\sigma_i) C_{ij} e^{-z\sigma_{ij}} \right) \right],\end{aligned}\quad (26)$$

$$D_{ij} = -a_j e^{z\sigma_{ij}}, \quad C_{ij} = -a_j e^{z\sigma_j/2} \left[\frac{\hat{B}_i e^{-z\sigma_i/2}}{z} - e^{z\sigma_i/2} \right]. \quad (27)$$

Because $q(r)$ is real, evaluating at $s = +iQ$ merely conjugates m and leaves (25) unchanged. Real symmetry and positive-definiteness of $\mathbf{S}^{\text{AL}}$ are *consequences* of this form rather than constraints imposed on it, and are therefore available as checks.

6 Charge polydispersity: $\delta_i \neq 1$ in closed form

Correcting eq. (22) does more than remove an error: it collapses what had looked like a substantial outstanding derivation.

With δ on the row index, eq. (21) becomes

$$C_{ij} + D_{ij} = \frac{2\pi}{z} \sum_k \rho_k \tilde{g}_{ik} D_{kj} = -\frac{2\pi}{z} a_j e^{z\sigma_j/2} \underbrace{\sum_k \rho_k \delta_k \tilde{g}_{ik} e^{z\sigma_k/2}}_{\equiv U_i} \quad (28)$$

so the contraction of $\tilde{g}$ that appears carries a δ *weight* — which is exactly what $\hat{B}$ supplies. By eq. (43) and the symmetry of $\tilde{g}$, $U_j = \hat{B}_j e^{-z\sigma_j/2}/2\pi$, and therefore

$$C_{ij} = \delta_i a_j e^{z\sigma_{ij}} - \frac{a_j}{z} e^{z\sigma_j/2} \hat{B}_i e^{-z\sigma_i/2}, \quad (29)$$

$$2\pi\sigma_{ij} g_{ij}(\sigma_{ij}^+) = A_j \frac{\sigma_i}{2} + \beta_j - z\delta_i a_j + a_j \hat{B}_i e^{-z\sigma_i}, \quad (30)$$

in closed form for *any* δ_i , reducing to (14) when $\delta_i = 1$. With eq. (22) as printed the same contraction would instead be unweighted, which $\hat{B}$ cannot supply, and C_{ij} would require solving eq. (21) as a coupled system — the obstacle that had made this case look hard. It was an artefact of the misprint.

Validated against the independent numerical OZ solver at $\delta_i \neq 1$: contact values agree to $1.0\text{--}3.6 \times 10^{-5}$ relative, $S_{ij}(Q)$ to $5\text{--}7 \times 10^{-5}$ absolute (both at that solver’s floor), with $g_{ij} = g_{ji}$ to $\leq 2 \times 10^{-16}$ and $\mathbf{S}^{\text{AL}}$ positive definite — neither imposed.

A second, structurally different route to the same result is implemented in `delta_general.py`, which obtains the contraction U_i not from $\hat{B}$ but from the Laplace-transform relations eq. (51)–eq. (53) and an $N \times N$ linear solve. It agrees with (30) to 4.4×10^{-16} at $N = 1$, exactly at $N = 2$ with $\delta = (1, 0.7)$, and 8.9×10^{-16} at $N = 3$ with $\delta = (1.3, 1, 0.6)$. Both routes assume the corrected eq. (22), so this is a check of the C_{ij} assembly and of the $\hat{B}$ shortcut under general δ , not independent evidence for the correction itself; that evidence is in §3.4 and Table 5.

Relation to earlier work. The factorisable coupling $K_{ij} = K\delta_i\delta_j$ is not new, and neither is its application to polydisperse colloids. Ginoza developed the factorisable multicomponent Yukawa MSA through the 1980s, and Ginoza & Yasutomi [8] built an analytical model of a colloidal dispersion carrying *simultaneous* size and “charge” polydispersity — their Z_i is our δ_i , and their $\phi_{ij}(r) = \varepsilon_0 Z_i Z_j (\sigma_0/r) e^{-z(r-\sigma_{ij})}$ is our potential — using a Schulz size distribution, eight years before BA. What they extract from it is thermodynamic: the Helmholtz free energy by a coupling-parameter integration, and the resulting equation of state and its critical behaviour as a function of the two polydispersity indices. They do not report partial structure factors, which is what a scattering application needs. The contribution here is therefore not the case $\delta_i \neq 1$ as such, but $S_{ij}(Q)$ and the contact matrix for it in closed form — and, as §11 shows, the observation that the RMSA rescaling is an instance of it.

7 Numerical implementation

Three issues proved to matter enough to record.

Auxiliary functions. The functions (7) are entire but their closed forms lose precision catastrophically for small $|x|$ — ψ_1 ’s numerator is $O(x^3/12)$ and is worthless below $|x| \sim 10^{-4}$. They are evaluated by Taylor series for $|x| < 0.5$ with 16 terms and by the closed form otherwise, and are complex-safe, since (25) needs imaginary arguments. An earlier version carried series branches for ϕ_1 and ψ_1 that contradicted their own closed forms — ϕ_1 ’s with the wrong sign — but with thresholds (10^{-6} , 10^{-5}) never reached for physical $z\sigma$; they were dormant, and no result predating the fix is affected. They are corrected because $Q \rightarrow 0$ in (25) does reach small $|x|$. All three now satisfy $\phi_1 = x\psi_1 - \phi_0/2$ to machine precision across the switch-over.

Reduced units. The closure is solved with lengths in units of $\sigma_{\text{ref}} = \langle \sigma^3 \rangle^{1/3}$. Feeding physical lengths (e.g. $\sigma \sim 1.0 \times 10^2 \text{ \AA}$) leaves the physics unchanged but moves a , $\hat{B}$ and Γ by many orders of magnitude; in particular Γ falls to $\sim 10^{-5}$, which a fixed search grid silently misses.

Root finding for Γ . Γ has dimensions of inverse length (it appears only as $z + \Gamma$), so its search grid is scaled by z and spaced geometrically over ten decades, rather than being absolute.

8 Validation

8.1 Single component against the compiled reference

At $N = 1$ the solution is compared with SASfit’s own compiled one-Yukawa library. The mapping between the two conventions was read off the C source rather than guessed; with $12\varphi \equiv 2\pi\rho$,

$$A = 2\pi a, \quad \beta = \pi a + 2\pi b, \quad C = 2\pi c, \quad D = 2\pi d, \quad (31)$$

under which BH02 eq. (21) becomes *identical* to the C source’s own nonlinear equation. Converting the library’s solution and solving our closure for the coupling that zeroes it returns $K_{\text{recovered}}/K = 1.000000$: there is no convention mismatch, and the two solvers find the same solution.

Table 6: Contact value at $N = 1$ against the reference library’s exact $(a + b) - cZe^{-Z}$. The last three rows have $K < 0$, i.e. repulsive.

Z	K	φ	reference	this work	rel. error
6.0	+1.0	0.10	2.131770741	2.131770741	8.3×10^{-16}
6.0	+0.5	0.10	1.708192190	1.708192190	3.9×10^{-16}
2.0	+0.3	0.10	1.498004160	1.498004160	1.5×10^{-16}
2.5	+0.5	0.10	1.654039213	1.654039213	2.7×10^{-16}
3.0	+1.0	0.30	2.666963463	2.666963463	4.2×10^{-15}
8.0	+2.0	0.15	3.113854120	3.113854120	1.1×10^{-14}
4.0	+0.8	0.25	2.373292365	2.373292365	9.4×10^{-16}
10.0	+3.0	0.35	4.528229745	4.528229745	6.3×10^{-14}
0.5	+0.1	0.05	1.212724935	1.212724935	3.1×10^{-15}
6.0	-1.0	0.10	0.503583757	0.503583757	4.4×10^{-15}
3.0	-1.0	0.20	1.289302599	1.289302599	1.0×10^{-15}
6.0	-2.0	0.10	-0.253002245	-0.253002245	2.2×10^{-14}

Agreement is at the level of 10^{-14} in the contact value (Table 6) and 10^{-10} in $S(Q)$ (Table 7), for both signs of K .

Table 7: $S(Q)$ at $N = 1$ against the reference library, worst relative error over 12 values of $Q \in [0.1, 400]$.

Z	K	φ	worst rel. error
6.0	+1.0	0.10	2.2×10^{-11}
2.0	+0.3	0.10	2.2×10^{-11}
3.0	+1.0	0.30	2.0×10^{-10}
10.0	+3.0	0.35	2.7×10^{-10}
6.0	-1.0	0.10	2.2×10^{-11}
3.0	-1.0	0.20	8.7×10^{-11}

A caveat on strong attraction. At $Z = 6$, $K = -2$, $\varphi = 0.1$ the contact value is *negative* (-0.253). Our solver reproduces the reference exactly there, so this is not an implementation error: it is the MSA itself becoming unphysical at strong attraction. Users should check $g_{ij}(\sigma_{ij}^+) > 0$ rather than assume a returned solution is physical.

8.2 The hard-sphere limit

As $K \rightarrow 0$ the contact value must reduce to the exact Percus–Yevick result $g(\sigma^+) = (1 + \varphi/2)/(1 - \varphi)^2$ for $N = 1$, and to the Lebowitz mixture values for $N > 1$. Both hold to $\sim 10^{-11}$ (Table 8), limited only by the $K \rightarrow 0$ extrapolation.

Table 8: $K \rightarrow 0$ hard-sphere limit at $N = 1$ ($K = 10^{-10}$).

φ	exact PY	this work	rel. error
0.05	1.1357340720	1.1357340721	8.0×10^{-11}
0.10	1.2962962963	1.2962962964	6.3×10^{-11}
0.20	1.7187500000	1.7187500001	3.7×10^{-11}
0.30	2.3469387755	2.3469387756	2.0×10^{-11}
0.40	3.3333333333	3.3333333334	9.5×10^{-12}

8.3 Multicomponent internal consistency

At $N > 1$ six independent conditions must hold *simultaneously*: the species-independence of Γ (eq. (72)), the componentwise closure (eq. (55)), the contracted closure (eq. (68)), the second contact-value expression (eq. (100)), the physical symmetry $g_{ij} = g_{ji}$ (eq. (77), which is nowhere imposed), and the large- Q tail of S_{ij} , which must match $4\pi\sqrt{\rho_i\rho_j}\sigma_{ij}g_{ij}$.

Before the corrections of §3.2 these were mutually *unsatisfiable*: any solution satisfying some violated others. After them, all six hold at machine precision (Table 9), flat across size ratios from 1 to 3.5.

Table 9: The six $N > 1$ conditions, before and after correcting eq. (35), eq. (38). Worst value over $N = 2$ ($\sigma = [1, 2]$ and $[1, 1.5]$) and $N = 3$ ($\sigma = [1, 1.5, 2.5]$).

Condition	before	after
Γ species-independence (eq. (72))	7.7×10^{-3} – 9.2×10^{-2}	$\leq 1.5 \times 10^{-16}$
componentwise closure (eq. (55))	10^{-16}	$\leq 2.2 \times 10^{-15}$
contracted closure (eq. (68))	10^{-17}	$\leq 5.6 \times 10^{-16}$
contact value (eq. (100))	1.7×10^{-3} – 1.0×10^{-2}	$\leq 2.4 \times 10^{-16}$
$g_{ij} = g_{ji}$ (eq. (77), not imposed)	—	$\leq 6.7 \times 10^{-16}$
off-diagonal $S_{i \neq j}$ tail	0.7%–2.5%	$\sim 10^{-6}$ (noise floor)

8.4 An externally independent test

All of the above is either internal to BH02’s equation set or ties S_{ij} back to our own contact values. Since four errors have been found in that equation set, an external check is warranted. We therefore wrote a numerical multicomponent Ornstein–Zernike solver that shares no equation with the analytic route: it iterates the matrix OZ relation on a radial grid under the same physical closure (2) and nothing else, with $\mathbf{S} = (\mathbf{I} - \hat{C})^{-1}$, Picard iteration on $\gamma = h - c$, and DST-I radial transforms.

Convergence is first order in the grid spacing — the hard-core discontinuity, not a defect — verified against the exact $N = 1$ answer, where the error ratio between $dr = 0.005$ and 0.0025 is 2.007, 2.003, 2.006, 1.999, 1.998 across five Q . Richardson extrapolation therefore applies and gains a factor ~ 20 , placing the method’s own floor at $\sim 4 \times 10^{-5}$ absolute in S and $\sim 10^{-4}$ relative in the contact value — well below the effects at issue.

Against this external reference (Table 10) the corrected solution sits *at* the numerical floor, while the as-printed equations lie 10–450 times above it. An entirely independent method therefore confirms §3.2 and rejects eq. (35), eq. (38) as printed.

Table 10: Relative error against the Richardson-extrapolated numerical OZ solution.

System	quantity	corrected	as printed
$N = 2, \sigma = [1, 2]$	contact values	$7.5 \times 10^{-5} - 1.5 \times 10^{-4}$	$1.5 \times 10^{-3} - 6.2 \times 10^{-3}$
$N = 3, \sigma = [1, 1.5, 2.5]$	contact values (worst of six)	5.8×10^{-5}	1.5×10^{-2}
$N = 2$	S_{00} (absolute)	$1.9 - 4.9 \times 10^{-5}$	up to 5.1×10^{-4}
$N = 2$	S_{01} (absolute)	$1.8 \times 10^{-6} - 4.7 \times 10^{-5}$	up to 4.0×10^{-4}

9 From a size distribution to the scattered intensity

9.1 Discretising the size distribution

A continuous distribution is mapped onto N classes (R_i, w_i) with $\sum_i w_i = 1$. Each distribution is integrated with *its own natural quadrature*: Gauss–Legendre in log space for the log-normal, generalised Gauss–Laguerre (with $\alpha = z$) for the Schulz–Zimm, the latter exact for every polynomial moment. This matters because $I(Q \rightarrow 0)$ depends on $\langle R^3 \rangle^2$ and $\langle R^6 \rangle$, so the highest moment is always the binding one.

The quadrature is truncated at $\pm \text{nsig}$ standard deviations, which trades truncation bias against numerical dynamic range: the S_{ij} assembly carries $e^{z\sigma_{ij}}$ factors, so an over-wide grid loses precision by cancellation. Untruncated Gauss–Hermite, though formally exact for the log-normal moments, places nodes out to $|u| \sim 6$ and at $\text{nbins} = 24$ spans R by a factor 165 with weights down to 10^{-17} and $e^{z\sigma} \sim 10^{33}$ — bins that hold no particles but wreck the assembly. Table 11 gives the trade-off; the default is $\text{nsig} = 5$, and a guard warns when $z\sigma_{\max} > 45$.

Table 11: Relative error of $\langle R^6 \rangle$ (the binding moment) against truncation width, and the resulting $z\sigma_{\max}$.

rel. s.d.	nsig=3	4	5	6	$z\sigma_{\max}$ at nsig=5
0.15	1.5×10^{-2}	9.0×10^{-4}	2.0×10^{-5}	7.7×10^{-6}	12
0.31	1.1×10^{-1}	1.4×10^{-2}	6.9×10^{-4}	9.1×10^{-6}	23
0.53	5.0×10^{-1}	1.6×10^{-1}	2.3×10^{-2}	1.4×10^{-3}	49 (guard fires)

Above rel. s.d. ≈ 0.4 there is no good choice: raising nsig to fix the moments pushes $z\sigma_{\max}$ past the point where the assembly holds precision. This is a genuine limitation of the approach, not a tuning problem.

Regarding nbins: $S(Q)$ and $I(Q)$ at low and intermediate Q converge to 5–6 digits by nbins ≈ 12 ; the deep form-factor minima, where I has fallen by $\sim 300\times$ and is a near-cancellation of oscillating F_i , need ≈ 20 .

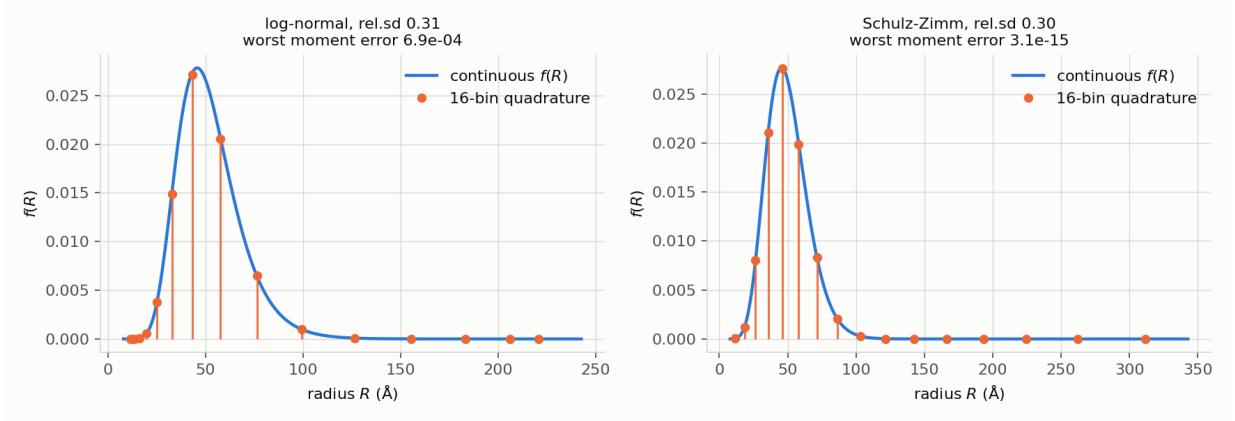


Figure 1: Continuous size distributions and their quadrature representation. Schulz-Zimm moments are reproduced to machine precision at any width; the log-normal to $\sim 10^{-4}$ at these settings.

9.2 Form factors and the intensity

Two form factors are provided: the homogeneous sphere, and a core-shell sphere polydisperse in the outer radius (with either a fixed shell thickness or a fixed core/shell ratio). The hard core and the Yukawa tail act on the *outer* diameter, so two systems differing only in internal contrast share identical $S_{ij}(Q)$ — structure and form factor remain separable, which is precisely what the decoupling approximation destroys.

The exact intensity is (1). Its validation is summarised in Table 12; note that in the first two rows the residual scales as relsd^2 and as φ respectively, i.e. it is the genuine physical effect being switched off, not numerical error.

Table 12: Validation of the scattering layer.

Check	Result
monodisperse limit, $\text{relsd} \rightarrow 0$, vs. reference $n F ^2 S(Q)$	3.1×10^{-5} at 10^{-3} ; 3.1×10^{-7} at 10^{-4}
dilute limit, $\varphi \rightarrow 0$, vs. $n\langle F^2 \rangle$	4.6×10^{-4} at $\varphi = 10^{-4}$; 4.6×10^{-6} at 10^{-6}
core-shell with zero core contrast $\equiv$ plain sphere	0 (exactly)
$\sum_{ij} \sqrt{x_i x_j} S_{ij} = S_{\text{number}}$	8.9×10^{-16}
nbins convergence at low/mid Q (nbins = 20)	$\sim 10^{-10}$

10 Assessment of SASfit’s approximations

10.1 The six schemes

SASfit implements six ways of folding a structure factor into a polydisperse intensity. They are transcribed here from SASfit’s own manual (`doc/manual/SASfit_ch3.tex`); the GUI exposes the first five as radio buttons, while the sixth is documented but not offered there. With n the total

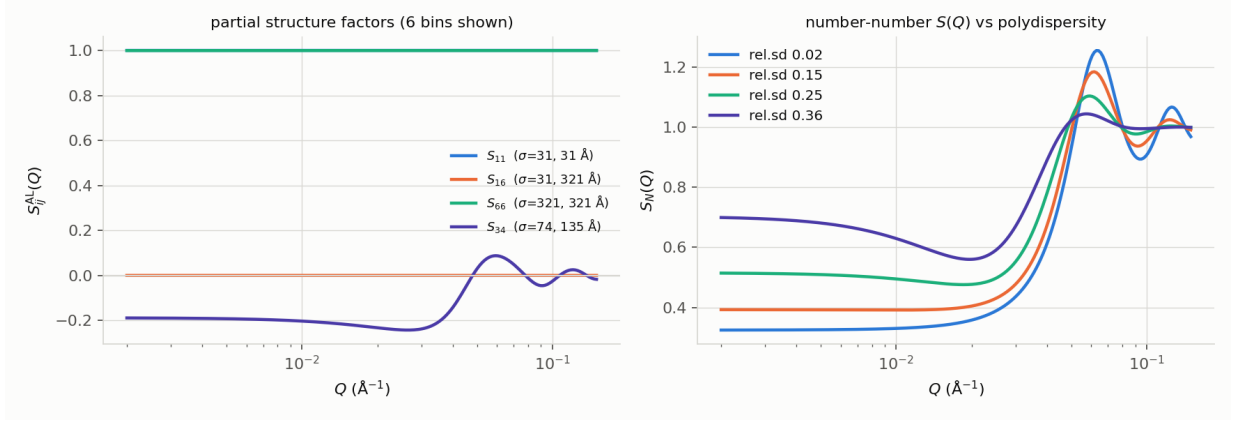


Figure 2: Left: partial structure factors for a six-class discretisation. Right: the number–number $S(Q)$ as polydispersity increases, at fixed volume fraction.

number density and w_i the number weights,

$$0 \text{ monodisperse: } I = n \langle F^2 \rangle S(Q) \quad (32)$$

$$1 \text{ decoupling [12]: } I = n \{ \langle F^2 \rangle + \langle F \rangle^2 [S(Q) - 1] \} \quad (33)$$

$$2 \text{ local monodisperse [13]: } I = n \sum_i w_i F_i^2 S(Q; R_i) \quad (34)$$

$$3 \text{ partial structure factors: } I = n \{ \langle F^2 \rangle + \sum_{ij} w_i w_j F_i F_j [S_{ij} - 1] \} \quad (35)$$

$$4 \text{ scaling [14]: as 3, each term} \times \bar{V}_{ij}/V_{\text{av}} \quad (36)$$

$$5 \text{ van der Waals one-fluid: as 4, normalised by } \langle \bar{V} \rangle \quad (37)$$

with $\bar{V}_{ij} = \frac{4\pi}{3} \left(\frac{R_i + R_j}{2} \right)^3$, $V_{\text{av}} = \langle V_i \rangle$ and $\langle \bar{V} \rangle = \sum_{ij} w_i w_j \bar{V}_{ij}$.

The essential point about schemes 3–5 is that their “ S_{ij} ” is *not* a true partial structure factor: it is the *monodisperse* $S(Q)$ evaluated at the mean radius $(R_i + R_j)/2$. That substitution is exactly what an exact S_{ij} removes, so the comparison below isolates its cost.

10.2 Method

Three interactions are used: an attractive Yukawa ($K = +1$), a repulsive Yukawa ($K = -1$; recall the sign convention of §2.1), and polydisperse adhesive hard spheres, for which the exact S_{ij} is taken from SASfit’s own Robertus engine [16] — a completely independent implementation, with no part of the Blum–Hernando machinery involved. All three run through *byte-identical* approximation code, so any difference between them is physics rather than implementation.

Errors are reported as the worst relative deviation from the exact result over the Q range carrying signal, defined as $I_{\text{exact}} > 1\%$ of its low- Q value. A plain maximum over all Q is not usable: $I(Q)$ spans decades and passes through form-factor minima where it nearly vanishes, so any ratio explodes there — and those minima are *deepest at low polydispersity*, producing a spurious error floor exactly where every scheme must become exact.

10.3 Results

10.4 Discussion

Three conclusions follow from Table 13 and Figure 4.

“Partial structure factors” is indistinguishable from decoupling. The two curves coincide at every polydispersity, in all three interactions. This is structural rather than accidental:

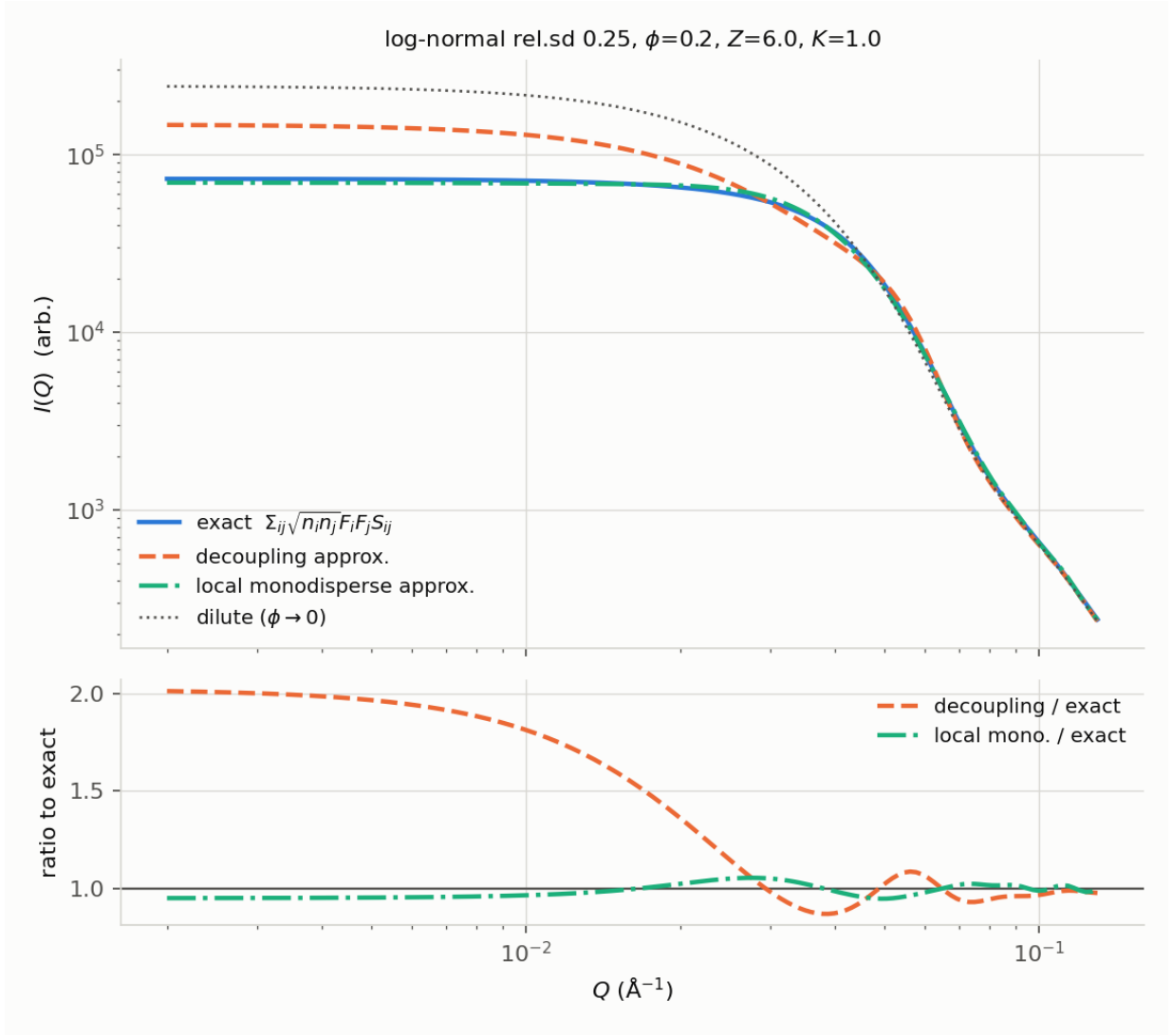


Figure 3: Exact $I(Q)$ against the decoupling and local monodisperse approximations, log-normal sizes with rel. s.d. 0.25 at $\phi = 0.20$. Lower panel: ratio to the exact result. Decoupling overestimates $I(Q \rightarrow 0)$ by a factor 2.0.

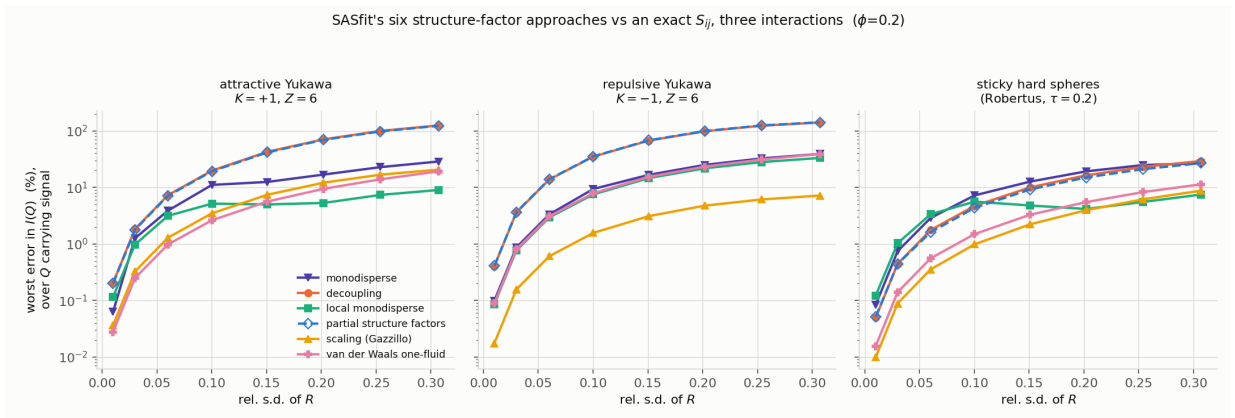


Figure 4: All six schemes against an exact S_{ij} , for three interactions. The “partial structure factors” curve (open diamonds, dashed) lies on top of the decoupling curve throughout. Note that the best-performing scheme changes between the left and centre panels.

Table 13: Worst error in $I(Q)$ (%) against the exact result, at $\varphi = 0.20$. Log-normal sizes; $Z = 6$ for the Yukawa cases, $\tau = 0.2$ for the adhesive spheres. Best scheme in each row in bold.

rel. s.d.	0 mono	1 decoup	2 LMA	3 partial	4 scaling	5 vdW1
<i>attractive Yukawa, $K = +1$</i>						
0.03	1.3	1.8	1.0	1.8	0.3	0.2
0.10	11.1	19.9	5.2	19.4	3.5	2.6
0.20	16.9	71.0	5.3	69.6	12.2	9.5
0.31	28.8	125.2	9.0	123.5	20.7	19.3
<i>repulsive Yukawa, $K = -1$</i>						
0.03	0.9	3.6	0.8	3.6	0.2	0.8
0.10	9.4	35.2	7.5	35.5	1.6	7.9
0.20	25.1	98.8	21.9	99.7	4.8	23.4
0.31	39.2	141.5	33.3	142.5	7.2	39.0
<i>adhesive hard spheres, $\tau = 0.2$ (Robertus)</i>						
0.03	0.7	0.4	1.0	0.4	0.1	0.1
0.10	7.3	4.7	5.6	4.4	1.0	1.5
0.20	19.3	16.4	4.2	15.0	4.0	5.5
0.31	26.8	29.4	7.5	27.0	8.8	11.4

scheme 3 substitutes the monodisperse S at the mean radius, discarding exactly what distinguishes a true S_{ij} , and so inherits decoupling’s failure mode intact. Since scheme 3 requires a double integral and scheme 1 does not, scheme 3 is strictly the more expensive way to obtain the same answer. This appears not to have been documented before.

The volume weighting is what does the work. Schemes 4 and 5 differ from scheme 3 *only* by the factor $\bar{V}_{ij}/V_{\text{av}}$ (or $/\langle\bar{V}\rangle$), and that single factor takes the error from $\sim 125\%$ to $\sim 20\%$ for the repulsive Yukawa. Gazzillo’s scaling argument is carrying real physical content.

The ranking depends on the interaction. For the *attractive* Yukawa the local monodisperse approximation is best (9% at rel. s.d. 0.31) and scaling is more than twice as bad (20.7%). Reversing the sign of K reverses the order: for the *repulsive* Yukawa, scaling becomes best by a wide margin (7.2%) while local monodisperse degrades to 33.3%. For adhesive spheres — an attractive interaction — the two are comparable, with scaling ahead at strong stickiness, consistent with the attractive Yukawa result.

This matters for practice because the commonest strongly interacting SANS system, a charge-stabilised colloid, is *repulsive*. The widely used guidance favouring the local monodisperse approximation is well founded for attractive systems, but for repulsive ones the scaling approximation of Gazzillo *et al.* is better by roughly a factor of four. This is a concrete, testable recommendation that was not previously checkable.

A structural caveat on the local monodisperse approximation. It can be *undefined* rather than merely inaccurate. It requires an auxiliary monodisperse solution at every class diameter, and the small- σ classes correspond to weak screening ($z\sigma_i \sim 1$) at the full volume fraction, where the one-Yukawa MSA has no physical root — even though the actual polydisperse system is perfectly well posed. Schemes 3–5 hit the same wall for the smallest *pairs*. The implementation reports the affected statistical weight and raises an exception rather than silently returning a curve computed from a subset of classes.

11 Extension to the rescaled MSA (RMSA)

The MSA returns $g_{ij}(\sigma_{ij}^+) < 0$ for strongly repulsive systems at low volume fraction — exactly the charge-stabilised colloids of interest in most SANS practice. The standard remedy is the rescaled MSA of Hansen and Hayter [15]: inflate the hard core, $\sigma \rightarrow \lambda\sigma$ with $\lambda > 1$, and fix λ by demanding that the rescaled system have $g(\text{contact}) = 0$ exactly.

RMSA is therefore *not* a different closure. It is the same MSA applied to a rescaled system, so the question is only whether the multicomponent solution admits the rescaling.

11.1 The rescaling maps onto $\delta_i \neq 1$

The physical potential is fixed; only the reference diameter in (2) changes. Holding $c_{ij}(r)$ fixed while re-referencing $\sigma_{ij} \rightarrow \lambda\sigma_{ij}$ gives $K'_{ij} = K_{ij}e^{-z(\lambda-1)\sigma_{ij}}$, and because $\sigma_{ij} = \frac{1}{2}(\sigma_i + \sigma_j)$ this *factorises*:

$$K'_{ij} = K \delta'_i \delta'_j, \quad \boxed{\delta'_i = \exp\left[-\frac{z}{2}(\lambda-1)\sigma_i\right]} \quad (38)$$

verified to reproduce $c_{ij}(r)$ to 5×10^{-16} . Two consequences follow.

Monodisperse: δ' is the same for every species, so it absorbs into the coupling, $K' = Ke^{-z(\lambda-1)\sigma}$, and the existing $\delta_i = 1$ machinery suffices. This is why Hansen–Hayter works as it does. Table 14 shows RMSA implemented directly on top of the present solver: the rescaling drives $g(\sigma'^+)$ to zero to $\sim 10^{-13}$ or better, with physically sensible λ between 1.03 and 1.30.

Polydisperse: δ'_i depends on σ_i and cannot be absorbed. Polydisperse RMSA is therefore *exactly* the $\delta_i \neq 1$ case of §12, item 1 — not a separate development.

Table 14: Monodisperse RMSA built on the present solver. g_{MSA} is the unrescaled contact value; λ is the rescaling factor found by requiring $g = 0$ in the rescaled system. Script: `rmsa_extension.py`.

φ	Z	K	g_{MSA}	λ	φ_{resc}	g_{resc}
0.02	2.0	-1.5	-0.2427	1.0639	0.0241	5×10^{-17}
0.02	2.0	-3.0	-1.4081	1.2673	0.0407	-3×10^{-13}
0.05	2.0	-3.0	-0.8372	1.1501	0.0761	-2×10^{-16}
0.10	2.0	-3.0	-0.1765	1.0285	0.1088	8×10^{-13}
0.02	1.0	-3.0	-1.0550	1.2997	0.0439	-6×10^{-16}
0.05	1.0	-3.0	-0.4274	1.1087	0.0682	-3×10^{-15}

11.2 Polydisperse RMSA is therefore analytic

Since §6 supplies $\delta_i \neq 1$ in closed form, the rescaled system can be solved directly and no numerical fallback is needed. We adopt a *common* rescaling factor for all species — a size-dependent λ_i would destroy the factorisation (38) and with it the analytic solution — and fix it by requiring every contact value to be non-negative, i.e. that the binding pair just reach zero:

$$\min_{ij} g_{ij}(\lambda\sigma_{ij}) = 0. \quad (39)$$

The number densities are physical and unchanged, so $\varphi \rightarrow \lambda^3\varphi$ follows automatically.

Table 15 shows the result. The condition is met to $\sim 10^{-14}$, every g_{ij} is then non-negative, and $\lambda = 1$ is returned unchanged when the plain MSA is already physical.

Cross-validation. For $\sigma = [1, 1.6]$, $\varphi = 0.05$, $Z = 2$, $K = -3$ the analytic route gives $\lambda = 1.224213$ and the independent numerical OZ solver, which shares no equation with it, gives 1.224143 — a difference of 0.006%.

Table 15: Polydisperse RMSA with a common λ fixed by (39). Number fractions 0.6 : 0.4 for $N = 2$ and 0.5 : 0.35 : 0.15 for $N = 3$. Script: `polydisperse_rmsa.py`.

σ	φ	Z	K	$\min g_{ij}$ (MSA)	λ	φ_{resc}
[1, 1.6]	0.02	2	-3	-1.6635	1.3110	0.0451
[1, 1.6]	0.05	2	-3	-1.2733	1.2242	0.0917
[1, 1.6]	0.10	2	-3	-0.7658	1.1212	0.1409
[1, 1.4, 2]	0.05	2	-3	-1.3478	1.2365	0.0945
[1, 1.6]	0.05	1	-3	-0.9367	1.2398	0.0953
[1, 1.6]	0.05	2	-0.5	+0.6768	1 (none needed)	0.0500

The small particles set the rescaling. In every case tested the binding pair is the smallest–smallest one. This follows directly from (38): $\delta'_i = e^{-z(\lambda-1)\sigma_i/2}$ is largest for the smallest σ_i , so K'_{11} is the strongest repulsion and g_{11} is the first contact value to go negative. The consequence is that a small minority population can force a substantial inflation of *every* diameter.

11.3 An unresolved question of principle

Condition (39) is a choice, not a derivation, and it is worth being explicit about what it costs. Monodisperse RMSA has one contact value and one rescaling parameter, so $g(\sigma^+) = 0$ determines λ uniquely. A mixture has $N(N+1)/2$ contact values and still only one λ , and they do not reach zero together: for the second row of Table 15, g_{11} vanishes at $\lambda = 1.2242$ while at that same λ the others sit at $g_{12} = 0.529$ and $g_{22} = 0.845$. Requiring the minimum to vanish repairs the worst-behaved pair and leaves the rest strictly positive — over-corrected relative to their own thresholds.

The alternatives each have a cost: a composition-weighted condition would leave some g_{ij} negative, defeating the purpose; species-dependent λ_i would break the factorisation (38), and with it the analytic solution, forcing the numerical route. We are not aware of a settled answer in the literature. It should be treated as a modelling decision made deliberately, not inherited from whichever convention an implementation happens to adopt.

12 Limitations and open items

1. **Charge polydispersity and polydisperse RMSA** — *resolved*, see §6 and §11. Both follow in closed form once eq. (22) is corrected. Listed here because earlier drafts of this work recorded them as the principal open item; the remaining uncertainty is the choice of rescaling condition (39), which is a modelling decision rather than a gap in the theory.
2. **Multiple Yukawa terms** ($M > 1$). Only $M = 1$ is implemented. The $M(M - 1)$ symmetry conditions of BH02 §4 become active for $M > 1$; the Γ -centric formulation of (11) was written with that generalisation in mind.
3. **Complex z_n** . Oscillatory closures are not supported.
4. **Very broad distributions** — *this item was wrong, and is withdrawn*. Earlier drafts recorded that above rel. s.d. ≈ 0.4 the quadrature could not simultaneously preserve $\langle R^6 \rangle$ and keep $z\sigma_{\text{max}}$ small enough for the S_{ij} assembly. That was inferred from the wrong diagnostic. A Gauss–Laguerre grid places its outermost nodes at enormous radii carrying no weight whatever — at rel. s.d. 0.5 and $N = 96$ the largest node sits at $z\sigma = 548$ with $w = 5 \times 10^{-151}$, a species containing no particles — so $z\sigma_{\text{max}}$ taken over all bins says nothing about the calculation’s health. The span over bins that actually hold particles is 0.2–17 and is essentially independent of N . Measured over $N = 12$ to 200, at nominal $z\sigma_{\text{max}}$ up to 548: the contact-matrix symmetry, the eq. (72) residual and the S_{ij} sum rule all remain at

10^{-15} , and $I(Q)$ converges cleanly (1.5×10^{-3} at $N = 12$, 1.2×10^{-9} at $N = 24$, 9×10^{-14} at $N = 48$ for rel. s.d. 0.2; 6×10^{-1} , 8.7×10^{-2} , 1.9×10^{-4} for rel. s.d. 0.5). Broad distributions are therefore not out of reach — they simply need more bins, and more bins work. $N = 200$ costs about 5 s. The guard now tests the weight-bearing span.

5. **Unphysical MSA solutions.** At strong attraction the MSA can return $g_{ij}(\sigma^+) < 0$. This is a property of the closure, not of the implementation, but callers should test for it.
6. **Size-independent stickiness.** The Robertus comparison inherits that engine’s assumption $\tau_{nm} = \tau$, so its polydispersity is purely in size — the same restriction as $\delta_i = 1$ on the Yukawa side.

13 Software

File	Role
<code>polydisperse_yukawa_msa.py</code>	closure solver; <code>solve()</code> and <code>solve_gamma()</code> ; contact values
<code>polydisperse_yukawa_sq.py</code>	$S_{ij}(Q)$ assembly, (25)
<code>polydisperse_sas_base.py</code>	exact $I(Q)$ and all six approximations (shared)
<code>polydisperse_yukawa_sas.py</code>	size distributions, form factors, Yukawa model
<code>robertus_shs_sas.py</code>	adhesive-sphere model on the same base
<code>robertusWrapper.py</code>	ctypes wrapper for SASfit’s <code>robertus_shs_core.c</code>
<code>polydisperse_rmsa.py</code>	polydisperse RMSA, common λ , condition (39)
<code>mixture_oz_numerical.py</code>	independent numerical multicomponent OZ solver
<code>sas_validation.py</code>	scattering-layer validation
<code>gamma_solve_validation.py</code>	the six $N > 1$ conditions
<code>external_N_validation.py</code>	numerical vs. corrected vs. as-printed
<code>PiX_two_routes.py</code>	eq. (34)/eq. (37) against eq. (30)/eq. (32)/eq. (33)
<code>Jhat_derivation_check.py</code>	derived vs. printed $\hat{\mathcal{J}}$; the three $P^{(n)}$
<code>eq100_gamma_check.py</code>	eq. (100)/eq. (100b) independent contact route
<code>blum_arias_IJ_check.py</code>	adjudicates eq. (35)/eq. (38) against BA eq. (76) (Table 4)
<code>eq22_symmetry_test.py</code>	decides the eq. (22) index by $g_{ij} = g_{ji}$ alone (Table 5)
<code>delta_general.py</code>	second route to general δ_i , via eq. (51)–eq. (53)
<code>contact_exact_verify.py</code>	contact value vs. the compiled library and the Lebowitz mixture limit
<code>rmsa_extension.py</code>	the rescaling algebra and monodisperse RMSA
<code>report_numbers.py</code>	regenerates every number in this report

Building the Robertus engine off Windows. It requires GSL and SUNDIALS ≥ 6 . The source targets SUNDIALS 7.x; on 6.x a small error-handler shim added with `-include` suffices, leaving the SASfit source itself unmodified:

```
gcc -O2 -fPIC -shared -include rshs/sundials6_shim.h -o librobertus.so \
    robertus_shs_core.c -lgsl -lgslcblas -lsundials_kinsol \
    -lsundials_nvecserial -lm
```

On Windows the wrapper locates SASfit’s prebuilt `libsasfit_robertus_shs.dll` automatically, or honours the `ROBERTUS_SHS_LIB` environment variable.

References

- [1] L. Blum and J. A. Hernando, *Yukawa fluids in the mean spherical approximation: I. The general solution*, J. Phys.: Condens. Matter **14**, 11933 (2002).

- [2] L. Blum and M. Arias, *Structure of multi-component/multi-Yukawa mixtures*, arXiv:cond-mat/0602477 (2006).
- [3] J. S. Høye and L. Blum, *Solution of the Yukawa closure of the Ornstein–Zernike equation*, J. Stat. Phys. **16**, 399 (1977).
- [4] L. Blum and J. S. Høye, *Solution of the Ornstein–Zernike equation with Yukawa closure for a mixture*, J. Stat. Phys. **19**, 317 (1978).
- [5] L. Blum, *Solution of the Ornstein–Zernike equation for a mixture of hard ions and Yukawa closure*, J. Stat. Phys. **22**, 661 (1980).
- [6] K. Niizeki, *Solution of the Ornstein–Zernike equation for a multicomponent system with Yukawa closure*, Mol. Phys. **43**, 251 (1981).
- [7] M. Ginoza and M. Yasutomi, *An extension of the analytical solution of the Ornstein–Zernike equation with the Yukawa closure*, J. Stat. Phys. **90**, 1475 (1998).
- [8] M. Ginoza and M. Yasutomi, *Analytical model for thermodynamic properties of a colloidal dispersion with size and ‘charge’ polydispersities*, Mol. Phys. **95**, 163 (1998).
- [9] Yu. V. Kalyuzhnyi and S. P. Hlushak, *Phase coexistence in polydisperse multi-Yukawa hard-sphere fluid: high temperature approximation*, J. Chem. Phys. **125**, 034501 (2006).
- [10] R. J. Baxter, *Percus–Yevick equation for hard spheres with surface adhesion*, J. Chem. Phys. **49**, 2770 (1968).
- [11] M. S. Wertheim, *Exact solution of the Percus–Yevick integral equation for hard spheres*, Phys. Rev. Lett. **10**, 321 (1963).
- [12] M. Kotlarchyk and S.-H. Chen, *Analysis of small angle neutron scattering spectra from polydisperse interacting colloids*, J. Chem. Phys. **79**, 2461 (1983).
- [13] J. S. Pedersen, *Determination of size distributions from small-angle scattering data for systems with effective hard-sphere interactions*, J. Appl. Cryst. **27**, 595 (1994).
- [14] D. Gazzillo, A. Giacometti, R. G. Della Valle, E. Venuti and F. Carsughi, *A scaling approximation for structure factors in the integral equation theory of polydisperse nonionic colloidal fluids*, J. Chem. Phys. **111**, 7636 (1999).
- [15] J.-P. Hansen and J. B. Hayter, *A rescaled MSA structure factor for dilute charged colloidal dispersions*, Mol. Phys. **46**, 651 (1982).
- [16] C. Robertus, W. H. Philipse, J. G. H. Joosten and Y. K. Levine, *Solution of the Percus–Yevick approximation of the multicomponent adhesive spheres system applied to the small angle x-ray scattering from a water-in-oil microemulsion*, J. Chem. Phys. **90**, 4482 (1989).